Supplementary Methods & Tables - nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

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Data Availability

The nf-core/viralmetagenome pipeline is freely available at `https://github.com/nf-core/viralmetagenome`. Scripts used to generate the results as well as parts of the results are available on github at `https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript`.

Supplementary Methods

S1. Evaluation of nf-core/viralmetagenome on public data

A public metagenomic dataset was constructed by selecting a random subset of 2,000 records from the NCBI Virus database, targeting the following viruses: *Orthonairovirus haemorrhagiae*, *Mammarenavirus lassaense*, *Zika virus*, *West Nile virus*, *Monkeypox virus*, *Influenza A virus*, *Severe acute respiratory syndrome coronavirus 2*, and *Human respiratory syncytial virus A*. Records were filtered to retain only those with associated SRA data, sequenced on the Illumina platform, and with a LibraryStrategy other than 'Amplicon'. From the resulting pool, 28 samples were randomly selected and downloaded using nf-core/fetchngs. Samples were then run with nf-core/viralmetagenome using default options with the exception of `-cluster_method 'mmseqs-cluster'` due to sequence size limitations of cdhit, and `-skip_read_classification` to speed up the analysis.

For plant pathogens, two samples were randomly selected from the SRA after querying for the following viruses: *Tobacco mosaic virus*, *Tomato spotted wilt virus*, *Cucumber mosaic virus*, and *Potato virus Y*. These were also downloaded using nf-core/fetchngs and processed with viralmetagenome using the same parameters (`-cluster_method 'mmseqs-cluster'`, `-skip_read_classification`), with the addition of the Virosaurus plant virus database as annotation database (`-annotation_db https://ftp.expasy.org/databases/viralzone/2020_4/virosaurus98_plant-20200330.fas.gz`) and RVDB's v29.0 unclustered database as reference pool (`-reference_pool https://rvdb.dbi.udel.edu/download/U-RVDBv29.0.fasta.gz`).

S2. Sample mixture resolution

To evaluate the pipeline's ability to distinguish viral genomes from the same species in mixed samples, we simulated a set of HIV-1 coinfections. We retrieved a random subset of 2,000 HIV-1 genomes from NCBI Virus (taxid: 11676) and selected MN090277.1 as the constant representative genome. We then calculated the pairwise Average Nucleotide Identity (ANI) between all 2,000 genomes and this representative using MMseqs2 (Steinegger and Söding 2017). Based on these ANI values, we binned the genomes into 0.5% intervals (ranging from 74% to 100%) and selected one target genome from each bin (n=38). Synthetic paired-end reads were generated for both the selected target genome and the representative genome using InSilicoSeq v2.0.1 (Gourlé et al. 2019) in 'kde' mode with the 'MiSeq' error model. Mixtures were simulated at a 50:50 abundance ratio with 50x coverage for each genome.

The simulated mixtures were analyzed using viralmetagenome with the parameters `-skip_preprocessing`, `-skip_read_classification`, and with U-RVDBv29.0 as a reference pool. We characterised the pipeline's performance by comparing the reconstructed consensus genomes to the original target genomes, defining three distinct performance zones based on empirical ANI thresholds (Supplementary figure 1):

- **Unresolved Zone (>88.7%):** At relatively high genetic similarity between target genomes in the mixture, the pipeline is unable to distinguish between the coinfecting strains, collapsing the mixture into a single consensus sequence per reference bin.
- **Ambiguous Zone (85.8% – 88.7%):** In this intermediate transition window, the pipeline successfully identifies multiple strains; however, the relatively high sequence similarity still permits reads to cross-map between references. This causes the reconstructed consensus sequences to accumulate mismatches in shared regions, resulting in lower fidelity to the original genomes despite initial separation by the assemblers.
- **High-Fidelity Resolution (< 85.8%):** Below this similarity threshold, the strains are sufficiently divergent to minimize cross-mapping interference, allowing for the accurate reconstruction of distinct, high-quality consensus genomes.

S3. Influence of scaffolding reference

The choice of scaffolding reference can considerably influence the completeness of the reconstructed genome. To quantify this effect, we analysed a subset of 7 Lassa virus samples from our own sequencing efforts (Lassa virus - Illumina Miseq data with UMIs). We compared consensus genomes generated using a gradient of scaffolding reference sequences ranging from 64% to 100% Average Nucleotide Identity (ANI) relative to the sample's baseline consensus. To establish this gradient, we first generated a baseline consensus for each sample using the unclustered RVDB (U-RVDBv29.0) as the reference pool. We then retrieved all *Mammarenavirus lassaense* (taxid: 3052310) sequences from NCBI and calculated their ANI against these baseline consensus genomes using MMseqs2 (Steinegger and Söding 2017). From this pool, we selected one scaffolding reference genome at every 2% ANI interval for each sample (Supplementary Figure 2). The Lassa virus samples exhibited

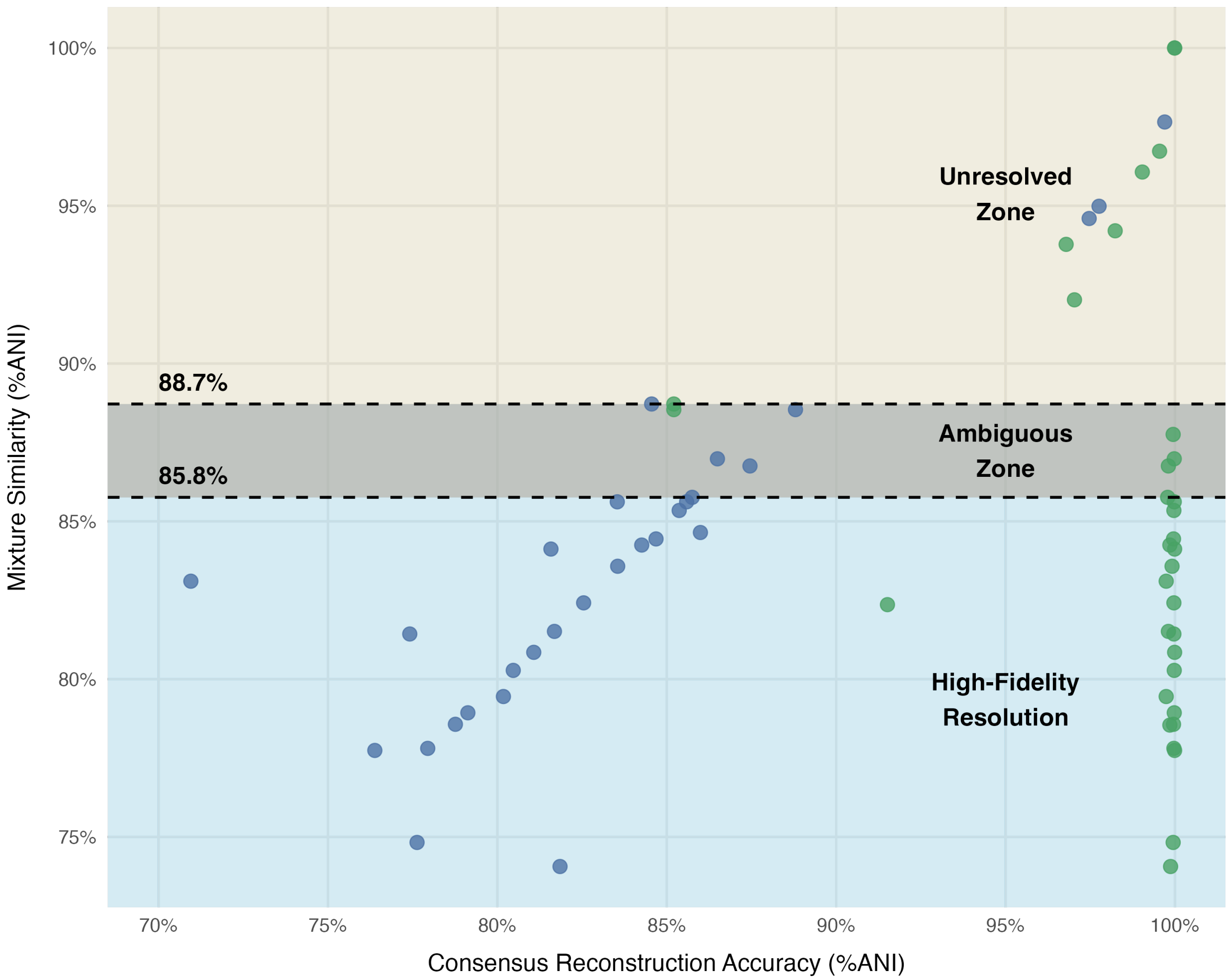

Figure 1: Mixture resolution dynamics across a genetic diversity gradient. This scatter plot characterises nf-core/viralmetagenome's ability to resolve coinfections of varying genetic similarity. The Y-axis represents the Average Nucleotide Identity (ANI) between the two target genomes present in the simulated mixture (Target vs. Representative), while the X-axis indicates the ANI of the reconstructed consensus genomes compared to the constant target genome.

differential coverage between segments (S segment mean 231X, L segment mean 91X), allowing us to evaluate scaffolding performance on both complete and fragmented assemblies.

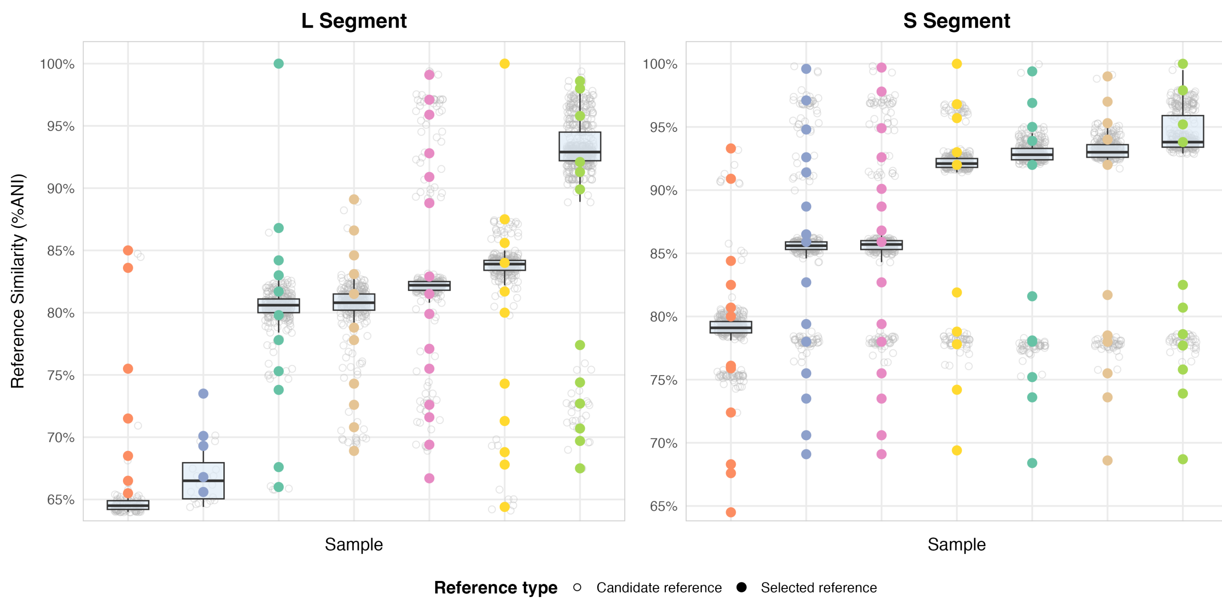

Figure 2: Jitter-boxplot for scaffolding reference genomes considered in the sample setup for determining the influence of the scaffolding reference. X-axis represents the samples used, y-axis represents the MMseqs ANI similarity towards the consensus genome. The consensus genome that is compared to all the references is the consensus genome generated by nf-core/viralmetagenome when automatic reference selection was applied for the U-RVDB reference library supplied with `-reference_pool`.

By default, nf-core/viralmetagenome selects the cluster centroid as a scaffolding reference. To assess the reference influence, the pipeline's code needed to be slightly modified to force NCBI genomes as scaffolding reference. This version of the code repository is available in the branch `https://github.com/Joon-Klaps/viralmetagenome/tree/force-external-ref-centroid`. The pipeline was then ran with the following parameters: `-with_umi`, `-umi_deduplicate both`, `-skip_read_classification`, `-skip_precluster`, `-intermediate_consensus_caller ivar`, `-consensus_caller ivar` and other default parameters.

To quantify the impact of the scaffolding reference on the final output, we compared the consensus genomes generated with the specifically supplied scaffold reference genomes against the consensus genomes generated when the U-RVDB was supplied (without scaffold reference forcing). We evaluated the sequence similarity as plotted in Figure 2, where we observe an almost linear correlation for the L segment. For the S segment, this pattern is less apparent as de novo assemblers could reconstruct near-complete genomes for most samples. In the L segment, where coverage was lower, nf-core/viralmetagenome excelled by connecting individual contigs to improve completeness.

Supplementary Tables

Table 1: Overview of computational tools and methods used in the nf-core/viralmetagenome pipeline. The pipeline integrates multiple bioinformatics tools across different stages of viral metagenomic analysis, from preprocessing to consensus calling. Each tool was selected for its specific capabilities and performance characteristics relevant to viral sequence analysis.

Stage	Step	Tool	Explanation	Reference
Pre-processing	Read trimming	fastp	A high-performance preprocessing solution built in C++ that consolidates multiple quality control operations into a single workflow. Fastp delivers enhanced processing speed compared to traditional alternatives, like Trimmomatic, while handling adapter removal, quality assessment, base correction, read and filtering. Fastp also features automated adapter detection capabilities for various Illumina sequencing protocols.	(Chen et al. 2018)
		Trimmomatic	A comprehensive preprocessing solution tailored for Illumina sequencing platforms with specialised paired-end read handling capabilities. Provides multiple pre-processing options, including adapter removal, quality-based trimming using sliding window approaches, and length-based filtering.	(Bolger et al. 2014)
	Complexity filtering	bbduk	A high-performance, multi-threaded tool designed to combine various data-quality-related operations, including trimming, filtering, and masking, into a single pass. It is particularly effective for removing host and other contaminant DNA using k-mer matching strategies, which is crucial for accurate metagenomic analysis.	(Bushnell 2022)
		PRINSEQ++	A C++ implementation that significantly improves upon its predecessor, prinseq-lite.pl, in terms of computational efficiency. It offers extensive quality control features, including filtering by length, GC content, quality scores, N content, and sequence complexity (entropy/DUST score).	(Cantu et al. 2019)

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Stage	**Step**	**Tool**	**Explanation**	**Reference**
Metagenomic diversity	Taxonomy classification	Kraken2	An advanced taxonomic classification system with an improved k-mer-based approach through optimised memory usage and enhanced processing speed. Incorporates protein-level search capabilities for improved detection of divergent viral sequences, with a library of pre-built RefSeq indexes on `https://benlangmead.github.io/aws-indexes/k2`	(Wood et al. 2019)
		Kaiju	A protein-level taxonomic classifier that employs exact matching algorithms on translated sequences using the Burrows-Wheeler transform. It can optionally allow amino acid substitutions and tends to have higher sensitivity and precision compared to k-mer-based classifiers (Kraken2). Kaiju is particularly effective for organisms with limited reference representation. A library of pre-built indexes, including RVDB, can be found on `https://bioinformatics-centre.github.io/kaiju/downloads.html`.	(Menzel et al. 2016)
Assembly & polishing	Assembly	SPAdes	A de Bruijn graph-based assembler designed to overcome challenges in complex sequencing data, such as non-uniform coverage (using multisized de Bruijn graphs) and variable insert sizes. The Spades suite contains multiple specialised modes rnaSPAdes, coronaSPAdes, metaSPAdes for extra flexibility.	(Meleshko et al. 2021)
		MEGAHIT	A succinct de Bruijn graph-based assembler specifically designed for large and complex metagenomic datasets. MEGAHIT has demonstrated the ability to generate larger assemblies with longer contig N50 and average contig length. Its relatively low memory requirements make it well-suited for handling large datasets.	(Li et al. 2016)

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Stage	**Step**	**Tool**	**Explanation**	**Reference**
		Trinity	A powerful modular transcriptome reconstruction tool, capable of fully reconstructing a large fraction of transcripts, including alternative splice isoforms and transcripts from recently duplicated genes, even in the absence of a reference genome. It partitions sequence data into individual de Bruijn graphs, processing each independently to extract full-length isoforms and resolve paralogous genes. Trinity's sensitivity is comparable to methods relying on genome alignments.	(Grabherr et al. 2011)
	Read alignment	BWAmem2	An optimized version of the BWA-MEM algorithm enhanced with Intel-specific acceleration feature, providing approximately a twofold speedup in alignment throughput over BWA-MEM while ensuring identical SAM output. It is highly efficient for aligning short reads to large reference genomes.	(Vasimuddin et al. 2019)
		Bowtie2	A fast and memory-efficient alignment tool for aligning sequencing reads to long reference sequences, such as mammalian genomes. Bowtie2 supports gapped, local, and paired-end alignment modes, combining high speed, sensitivity, and accuracy	(Langmead et al. 2019)
Clustering		CD-HIT-EST	A highly efficient and widely used program for clustering large sets of protein or nucleotide sequences. Implementation constraints limit processing to genomes under 10Mbp with minimum 0.8. The program employs a greedy incremental algorithm.	(Li and Godzik 2006)
		MMSeqs	A powerful software suite for fast and sensitive deep clustering and searching of large protein sequence sets. It is significantly faster than other tools like BLASTclust and can cluster large databases down to low sequence identity thresholds. MMseqs offers a variety of clustering modes for a lot of flexibility.	(Steinegger and Söding 2017)
		VSEARCH	An open-source, versatile alternative for USEARCH. VSEARCH supports processing very large datasets, limited primarily by available memory.	(Rognes et al. 2016)

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Stage	Step	Tool	Explanation	Reference
		Mash	A non-alignment based clustering technique that applies MinHash algorithms for rapid genome and metagenome distance estimation. It compresses large sequences into small, representative sketches, enabling ultra-fast estimations of global mutation distances. Mash creates compact sequence representations enabling ultra-fast similarity assessments and large-scale clustering operations, though accuracy decreases below 95% genome similarity.	(Ondov et al. 2019)
		vRhyme	A machine learning-based binning solution specifically developed for viral genome recovery from metagenomic datasets. VRhyme addresses the unique challenges of viral sequences, such as the lack of universal marker genes. Addresses unique viral sequence characteristics through supervised learning approaches combined with coverage-based analysis across multiple samples. Generates high-quality viral bins with minimal computational overhead, though it may produce empty results when insufficient viral evidence is detected.	(Kieft et al. 2022)
Variant & consensus calling	Variant & consensus calling	iVar	A computational package specifically designed for viral amplicon-based sequencing, integrating functions like consensus and variant calling (including iSNVs and insertions/deletions). It is a key component of best-practice pipelines for reconstructing consensus genomes from viral sequencing data.	(Grubaugh et al. 2019)
		BCFtools	A robust suite of utilities for manipulating variant calls in VCF and BCF formats. Provides extensive functionality for variant calling using multiallelic models, data filtering, file merging, and consensus sequence generation through variant application.	(Danecek et al. 2021)

References

Anthony M Bolger, Marc Lohse, and Bjoern Usadel. Trimmomatic: a flexible trimmer for illumina sequence data. *Bioinformatics*, 30(15):2114–2120, 2014. doi: 10.1093/bioinformatics/btu170.

Brian Bushnell. BBMap, 2022.

Vito Adrian Cantu, Jeffrey Sadural, and Robert Edwards. PRINSEQ++, a multi-threaded tool for fast and efficient quality control and preprocessing of sequencing datasets. *PeerJ*, 2019. doi: 10.7287/peerj.preprints.27553v1.

Shifu Chen, Yanqing Zhou, Yaru Chen, and Jia Gu. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17):i884–i890, 2018. doi: 10.1093/bioinformatics/bty560.

Petr Danecek, James K Bonfield, Jennifer Liddle, John Marshall, Valeriu Ohan, Martin O Pollard, Andrew Whitwham, Thomas Keane, Shane A McCarthy, Robert M Davies, and Heng Li. Twelve years of SAMtools and BCFtools. *Gigascience*, 10(2), 2021. doi: 10.1093/gigascience/giab008.

Hadrien Gourlé, Oskar Karlsson-Lindsjö, Juliette Hayer, and Erik Bongcam-Rudloff. Simulating illumina metagenomic data with InSilicoSeq. *Bioinformatics*, 35(3):521–522, 2019. doi: 10.1093/bioinformatics/bty630.

Manfred G Grabherr, Brian J Haas, Moran Yassour, Joshua Z Levin, Dawn A Thompson, Ido Amit, Xian Adiconis, Lin Fan, Raktima Raychowdhury, Qiandong Zeng, Zehua Chen, Evan Mauceli, Nir Hacohen, Andreas Gnirke, Nicholas Rhind, Federica di Palma, Bruce W Birren, Chad Nusbaum, Kerstin Lindblad-Toh, Nir Friedman, and Aviv Regev. Full-length transcriptome assembly from RNA-seq data without a reference genome. *Nat. Biotechnol.*, 29(7):644–652, 2011. doi: 10.1038/nbt.1883.

Nathan D Grubaugh, Karthik Gangavarapu, Joshua Quick, Nathaniel L Matteson, Jaqueline Goes De Jesus, Bradley J Main, Amanda L Tan, Lauren M Paul, Doug E Brackney, Saran Grewal, Nikos Gurfield, Koen K A Van Rompay, Sharon Isern, Scott F Michael, Lark L Coffey, Nicholas J Loman, and Kristian G Andersen. An amplicon-based sequencing framework for accurately measuring intrahost virus diversity using PrimalSeq and iVar. *Genome Biol.*, 20(1):8, 2019. doi: 10.1186/s13059-018-1618-7.

Kristopher Kieft, Alyssa Adams, Rauf Salamzade, Lindsay Kalan, and Karthik Anantharaman. vRhyme enables binning of viral genomes from metagenomes. *Nucleic Acids Res.*, 50(14):e83, 2022. doi: 10.1093/nar/gkac341.

Ben Langmead, Christopher Wilks, Valentin Antonescu, and Rone Charles. Scaling read aligners to hundreds of threads on general-purpose processors. *Bioinformatics*, 35(3):421–432, 2019. doi: 10.1093/bioinformatics/bty648.

Dinghua Li, Ruibang Luo, Chi-Man Liu, Chi-Ming Leung, Hing-Fung Ting, Kunihiko Sadakane, Hiroshi Yamashita, and Tak-Wah Lam. MEGAHIT v1.0: A fast and scalable metagenome assembler driven by advanced methodologies and community practices. *Methods*, 102:3–11, 2016. doi: 10.1016/j.ymeth.2016.02.020.

Weizhong Li and Adam Godzik. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics*, 22(13):1658–1659, 2006. doi: 10.1093/bioinformatics/btl158.

Dmitry Meleshko, Iman Hajirasouliha, and Anton Korobeynikov. coronaSPAdes: from biosynthetic gene clusters to RNA viral assemblies. *Bioinformatics*, 2021. doi: 10.1093/bioinformatics/btab597.

Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. *Nat. Commun.*, 7:11257, 2016. doi: 10.1038/ncomms11257.

Brian D Ondov, Gabriel J Starrett, Anna Sappington, Aleksandra Kostic, Sergey Koren, Christopher B Buck, and Adam M Phillippy. Mash screen: high-throughput sequence containment estimation for genome discovery. *Genome Biol.*, 20(1):232, 2019. doi: 10.1186/s13059-019-1841-x.

Torbjørn Rognes, Tomáš Flouri, Ben Nichols, Christopher Quince, and Frédéric Mahé. VSEARCH: a versatile open source tool for metagenomics. *PeerJ*, 4:e2584, 2016. doi: 10.7717/peerj.2584.

Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.*, 35(11):1026–1028, 2017. doi: 10.1038/nbt.3988.

Md Vasimuddin, Sanchit Misra, Heng Li, and Srinivas Aluru. Efficient architecture-aware acceleration of BWA-MEM for multicore systems. In *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 314–324. IEEE, 2019. doi: 10.1109/IPDPS.2019.00041.

Derrick E Wood, Jennifer Lu, and Ben Langmead. Improved metagenomic analysis with kraken 2. *Genome Biol.*, 20(1):257, 2019. doi: 10.1186/s13059-019-1891-0.